Chapter 13: Meiosis and Sexual Life Cycles

Complete Master Reference Edition • Fused Curriculum (Introduction, Meiosis & Sexual Reproduction, The Process of Meiosis, Lecture Deck Synthesis, Vocabulary & Practice Problems)

Course: BSC2010C General Biology I **Institution:** University of Central Florida **Curriculum Block:** Block 4 (Exam 4

Section 13.1: Content Introduction & Hereditary Foundations

Biological inheritance, transmission of traits from parents to offspring, homologous chromosomes, autosomes vs. sex chromosomes, and diploid vs. haploid genomes.

Chapter 13 Content Introduction

Learning Objectives

- Explain how Meiosis creates haploid gametes
- Explain why sexual life cycles require Meiosis
- Describe and recognize the phases of Meiosis I and Meiosis II

Chapter 13: Meiosis and Sexual Life Cycles

Some organisms reproduce via asexual reproduction through the process of mitosis. In asexual reproduction there is only one parent who produces an offspring that is a clone; both parent and offspring are genetically identical. Given that the parent and offspring are genetically identical, you can easily see that there is no genetic variation between parent and offspring. Any genetic variation arises through random mutation.

Most people are more familiar with sexual reproduction. Sexually reproducing organisms reproduce via process of meiosis. In sexual reproduction there are two parents who each contribute half of their offspring's genetic make-up. In other words, offspring receive half of their chromosomes from their mother and half of their chromosomes from their father. This means that there is quite a bit of genetic variation and the offspring are never genetically identical to their parents. The more genetic variation, the more likely a species will be able to evolve.

We know that mitosis is a type of cellular division that maintains the diploid number. For example, if the parent cell had 10 chromosomes then the daughter cell will also have 10 chromosomes. Meiosis is a different type of cellular division that reduces the chromosome number from diploid to haploid. Meiosis only occurs in testes and ovaries to create haploid gametes (sperm and eggs). You know that humans have 46 chromosomes. In human males, meiosis produces haploid sperm that have 23 chromosomes. In human females, meiosis produces haploid eggs that have 23 chromosomes. Why is this necessary? What happens during fertilization? One sperm combines with one egg to form the very first cell of an offspring. One human sperm, with 23 chromosomes, combines with one human egg, with 23 chromosomes, such that the resulting cell (the first cell of the human zygote) has 46 chromosomes. Voila! If sperm and eggs each had 46 chromosomes then the very first cell of that human zygote would have 92 chromosomes...obviously that is not correct for humans.

Interactive Course Module: Haploid vs. Diploid Materia widget

CHAPTER READING

Chapter 13 Textbook Reading: Meiosis and Sexual Reproduction, The Process of Meiosis

CHAPTER LINKS

Chapter 13 PowerPoint: Link

Chapter 13 Post-Quiz

Chapter 12 and 13 Simple Practice Problems

Section 13.2: Meiosis and Sexual Reproduction

Evolutionary trade-offs of asexual reproduction (clones, rapid propagation, zero genetic variation) vs. sexual reproduction (gametes, energetic cost, immense genetic diversity).

Chapter 13 Meiosis and Sexual Reproduction

Learning Objectives

- Explain that meiosis and sexual reproduction are highly evolved traits
- Identify variation among offspring as a potential evolutionary advantage of sexual reproduction

The ability to reproduce is a basic characteristic of all organisms: Hippopotamuses give birth to hippopotamus calves; maple trees produce seeds from which maple tree seedlings emerge; and adult flamingos lay eggs that hatch into flamingo chicks. However, unlike these examples, offspring may or may not resemble their parents. For example, in the case of most insects such as butterflies (with a complete metamorphosis), the larval forms rarely resemble the adult forms.

Although many unicellular organisms and a few multicellular organisms can produce genetically identical clones of themselves through ***asexual reproduction***, most multicellular organisms reproduce regularly using another method—***sexual reproduction***. This highly evolved method involves the production, by the parents, of two haploid cells and the fusion of two haploid cells to form a single, genetically recombined diploid cell, a zygote—which is a genetically unique organism. In almost all sexually reproducing species, these two haploid cells differ in size, with the smaller cell called “male” and the larger one called “female." These haploid cells are produced by a type of cell division called meiosis. Sexual reproduction, involving both meiosis and fertilization, introduces variation into offspring that may account for the evolutionary success of sexual reproduction.

The zygote formed by fertilization of the egg by the sperm, undergoes thousands of rounds of mitotic cell division to eventually develop into an adult organism.

Sexual reproduction was likely an early evolutionary innovation after the appearance of eukaryotic cells. It appears to have been very successful because most eukaryotes are able to reproduce sexually and, in many animals, it is the only mode of reproduction. And yet, scientists also recognize some real disadvantages to sexual reproduction. On the surface, creating offspring that are genetic clones of the parent appears to be a better system. If the parent organism is successfully occupying a habitat, offspring with the same traits should be

similarly successful. There is also the obvious benefit to an organism that can produce offspring whenever circumstances are favorable by asexual budding, fragmentation, or by producing eggs asexually. These methods of reproduction do not require a partner with which to reproduce. Indeed, some organisms that lead a solitary lifestyle have retained the ability to reproduce asexually. In addition, in asexual populations, every individual is capable of reproduction. In sexual populations, the males are not producing the offspring themselves, so hypothetically an asexual population could grow twice as fast.

However, multicellular organisms that exclusively depend on asexual reproduction are exceedingly rare. Why are meiosis and sexual reproductive strategies so common? These are important (and as yet unanswered) questions in biology, even though they have been the focus of much research beginning in the latter half of the 20th century. There are several possible explanations, one of which is that the variation that sexual reproduction creates among offspring is very important to the survival and reproduction of the population. Thus, on average, a sexually reproducing population will leave more descendants than an otherwise similar asexually reproducing population. The only source of variation in asexual organisms is mutation. Mutations that take place during the formation of germ cell lines are also a source of variation in sexually reproducing organisms. However, in contrast to mutation during asexual reproduction, the mutations during sexual reproduction can be continually reshuffled from one generation to the next when different parents combine their unique genomes and the genes are mixed into different combinations by crossovers during prophase I and random assortment at metaphase I.

Section 13.3: The Two-Stage Process of Meiosis

Meiosis I reductional division (Prophase I synapsis, crossing over at chiasmata, Metaphase I independent assortment, Anaphase I homologous segregation) and Meiosis II equational division (sister chromatid separation).

Chapter 13 The Process of Meiosis

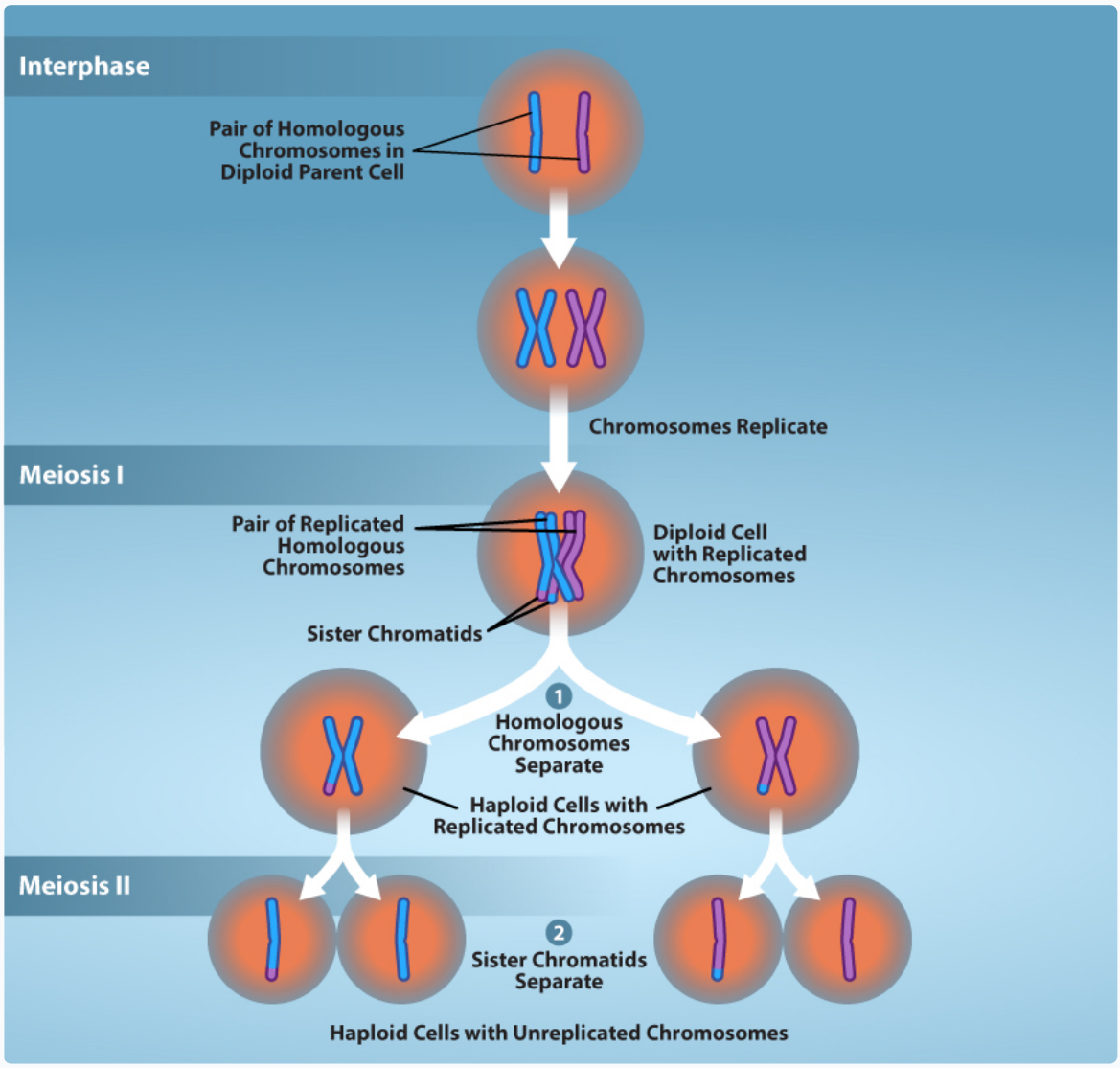
Learning Objectives

- Describe the behavior of chromosomes during meiosis, and the differences between the first and second meiotic divisions
- Describe the cellular events that take place during meiosis
- Explain the differences between meiosis and mitosis
- Explain the mechanisms within the meiotic process that produce genetic variation among the haploid gametes

Sexual reproduction requires the union of two specialized cells, called gametes (egg and sperm), each of which contains one set of chromosomes. When gametes unite, they form a one celled **zygote** that contains two sets of chromosomes; one haploid set from the sperm and one haploid set from the egg. For sexual reproduction to occur, specialized diploid cells within the individual must reduce the number of chromosome sets to produce haploid gametes; otherwise, the number of chromosome sets will double with every future round of fertilization. Therefore, sexual reproduction requires nuclear division that reduces the number of chromosome sets by half.

Animals and most plants are diploid and therefore have two sets of chromosomes. In each somatic cell of the organism (all cells of a multicellular organism except the gametes or reproductive cells), the nucleus contains two copies of each chromosome, called **homologous chromosomes**. Homologous chromosomes are matched pairs containing the same genes (but not necessarily the same version of those genes) in identical locations along their lengths. Diploid organisms inherit one copy of each homologous chromosome from the mother and one copy of each homologous chromosome from the father.

Meiosis is the *nuclear division* that forms haploid cells from diploid cells, and it employs many of the same cellular mechanisms as mitosis. However, as you have learned, **mitosis** produces daughter cells that are genetically identical to the original parent cell. In mitosis, both the parent and the daughter cells are at the same "ploidy level"—diploid. In meiosis, the parent cell is diploid and the daughter cells that result are haploid. To achieve this reduction in chromosome number, meiosis consists of one round of chromosome replication followed by two rounds of nuclear division. Because many events that occur during each of the division stages are analogous to the events of mitosis, the same stage names are assigned. However, because there are two rounds of division, the major process and the stages are designated with a "I" or a "II." Thus, meiosis I is the first round of meiotic division and consists of prophase I, metaphase I, and so on. Likewise, Meiosis II (during which the second round of meiotic division takes place) includes prophase II, metaphase II, and so on.



Three phases of meiosis are shown. In interphase, a pair of chromosomes replicate, resulting in a diploid cell with replicated chromosomes. In Meiosis 1, the chromosomes separate into haploid cells. In Meiosis 2, the sister chromosomes sepa...

Figure 11.2 Overview of Meiosis. The production of gametes is a crucial process for sexually reproducing organisms. Meiosis is the mechanism used to reduce diploid cells to haploid gametes while introducing genetic diversity. Prior to meiosis, chromosomes are replicated in S-phase to ensure proper number of chromosomes in the resulting gametes. During meiosis, two successive rounds of division reduces the number of chromosomes

(ploidy) of the cell by half. going from diploid cells to haploid gametes. Credit: Rao, A., Tag, A, Fletcher, S., and Ryan, K. Department of Biology, Texas A&M University.

Meiosis I

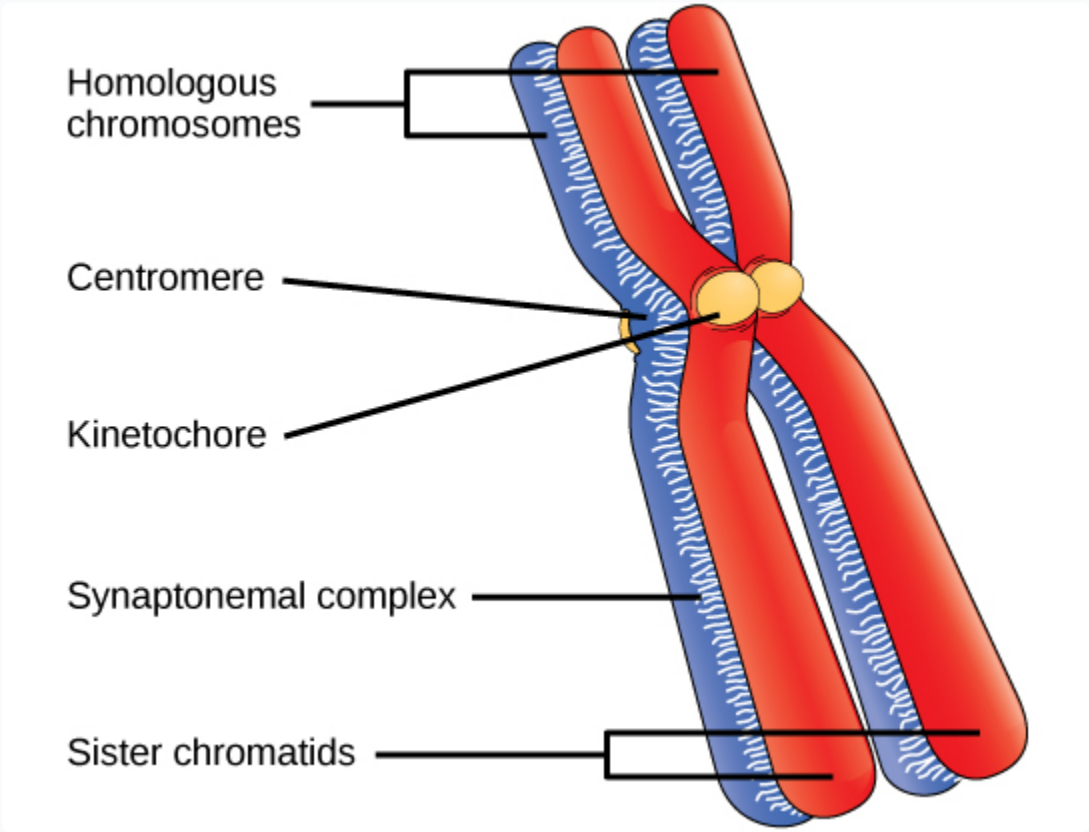
Meiosis is preceded by an interphase consisting of G_1, S, and G_2 phases, which are identical to the interphase phases preceding mitosis. The G_1 phase (the "first gap phase") is focused on cell growth. During the S phase—the second phase of interphase—the cell copies or *replicates* the DNA of the chromosomes. Finally, in the G_2 phase (the "second gap phase") the cell undergoes the final preparations for meiosis.

During DNA duplication in the S phase, each chromosome is replicated to produce two identical copies—*sister chromatids* that are held together at the centromere by cohesin proteins, which hold the chromatids together until anaphase II.

Prophase I

Early in prophase I, before the chromosomes can be seen clearly with a microscope, the homologous chromosomes come together to be physically attached to each other; this physical pairing of the homologous chromosomes is called ***synapsis***. In synapsis, the genes on the chromatids of the homologous chromosomes are aligned precisely with each other. The synapsed chromosomes exchange small sections of chromosomal segments between homologous nonsister chromatids—a process called **crossing over**. Crossing over can be observed visually after the exchange as *chiasmata* (singular = chiasma) (Figure 11.3). Crossing over creates recombinant chromosomes -

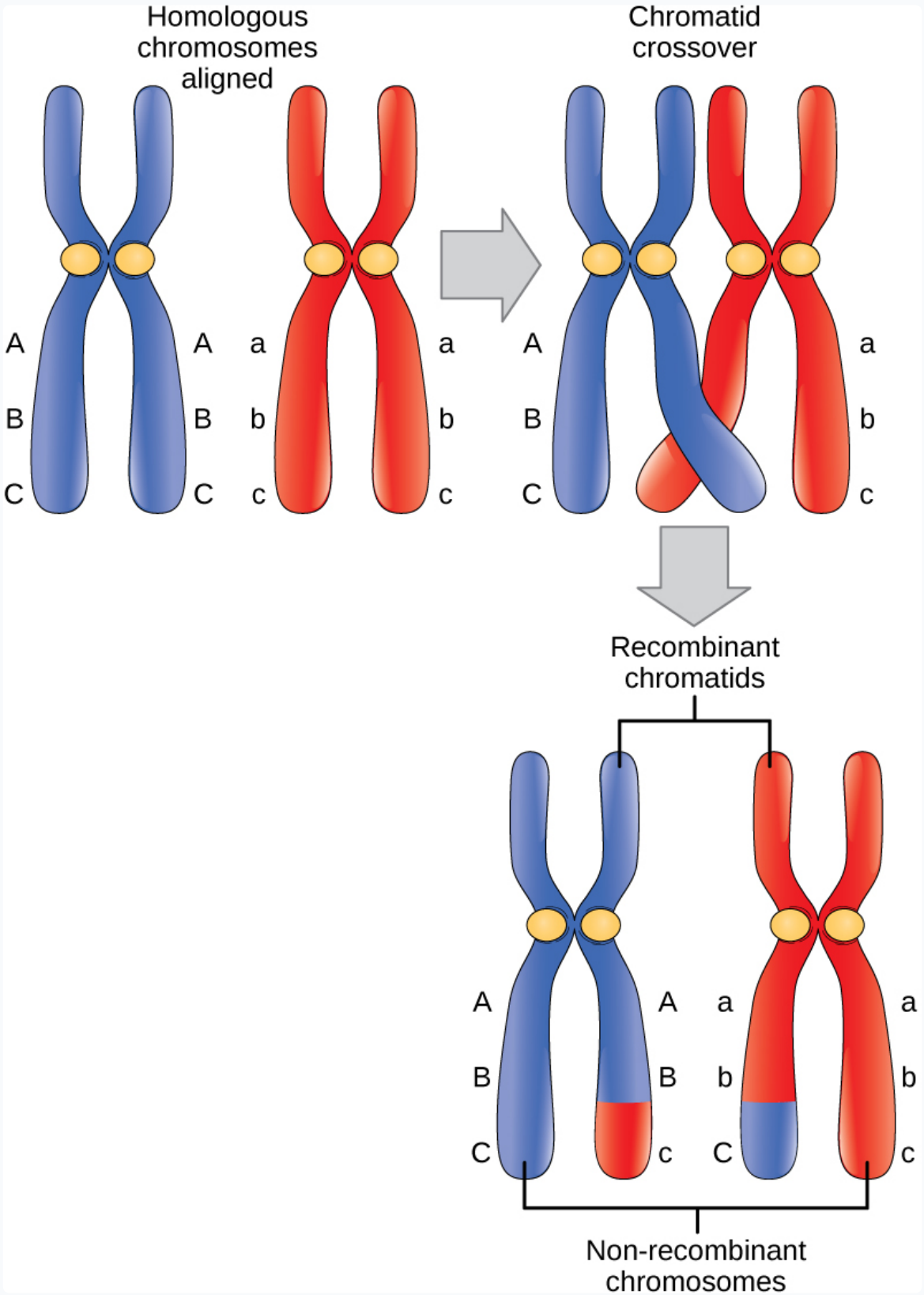
In humans, even though the X and Y sex chromosomes are not completely homologous (that is, most of their genes differ), there is a small region of homology that allows the X and Y chromosomes to partially pair up during prophase I.



This illustration depicts two pairs of sister chromatids joined together to form homologous chromosomes. The chromatids are pinched together at the centromere and held together by the kinetochore. A protein lattice called a synaptonemal com...

Figure 11.3 Early in prophase I, homologous chromosomes come together to form a synapse. The chromosomes are bound tightly together and in perfect alignment by a protein lattice called a synaptonemal complex and by cohesin proteins at the centromere.

The crossing over events are the first source of genetic variation in the nuclei produced by meiosis. A single crossover event between homologous nonsister chromatids leads to a reciprocal exchange of equivalent DNA between synapsed chromosomes. These chromosomes are called **recombinant chromosomes**. A recombinant contains a combination of maternal and paternal genes that did not exist before the crossover. Crossover events can occur almost anywhere along the length of the synapsed chromosomes. Different cells undergoing meiosis will therefore produce different recombinant chromatids, with varying combinations of maternal and parental genes. Multiple crossovers in an arm of the chromosome have the same effect, exchanging segments of DNA to produce genetically recombined chromosomes.



This illustration shows a pair of homologous chromosomes that are aligned. The ends of two nonsister chromatids of the homologous chromosomes cross over, and genetic material is exchanged. The non-sister chromatids between which genetic mat...

Figure 11.4 Crossover occurs between nonsister chromatids of homologous chromosomes. The result is an exchange of genetic material between homologous chromosomes.

Late in Prophase I, microtubules from each pole move toward the middle of the cell and attach to one of the kinetochores of the two fused homologous chromosomes. Each member of the homologous pair attaches to a microtubule extending from opposite poles of the cell so that in the next phase, the microtubules can pull the homologous pair apart. The homologous chromosomes are still synapsed. In addition, the nuclear membrane has broken down entirely.

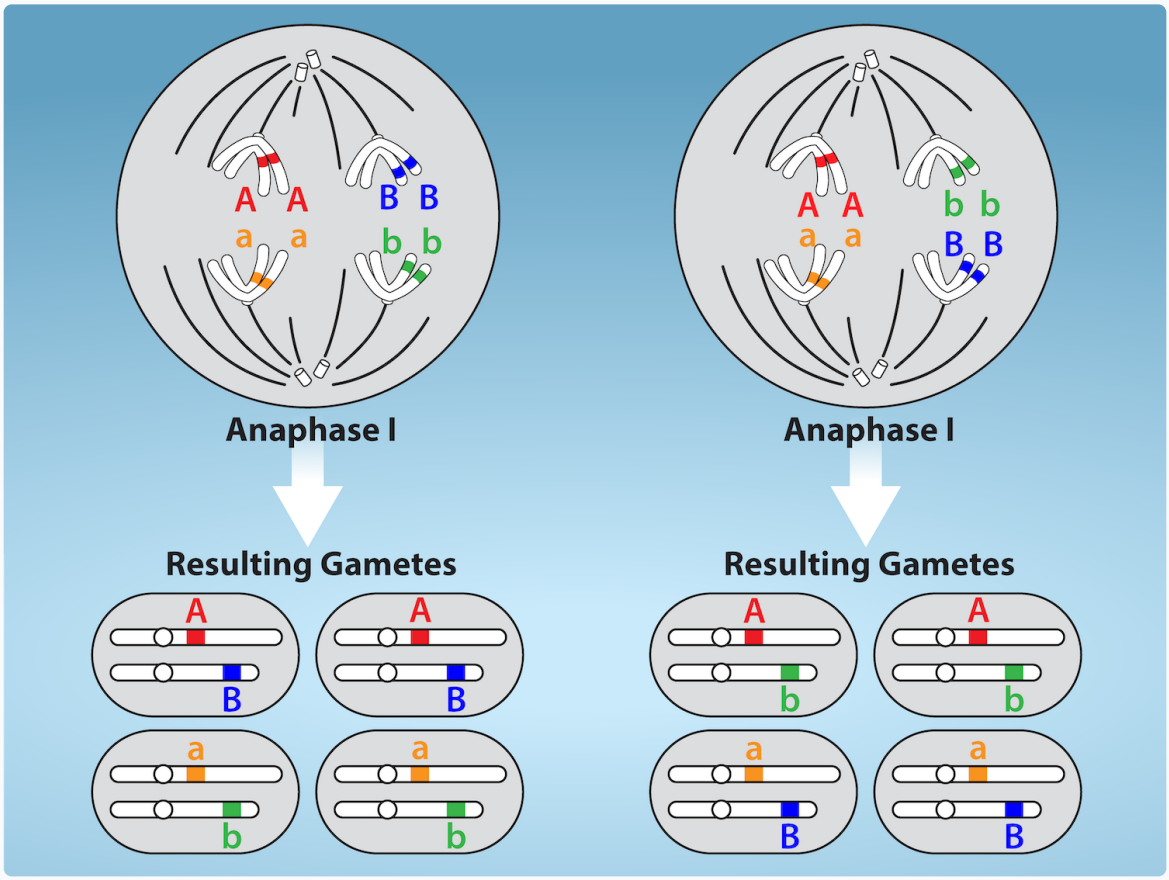
Metaphase I

During metaphase I, the homologous chromosomes are arranged at the **metaphase plate**— roughly in the midline of the cell, with the kinetochores facing opposite poles. Each homologous pair alignes randomly at the equator. For example, if the two homologous members of chromosome 1 are labeled *a* and *b*, then the chromosomes could line up a-b or b-a. This is important in determining the genes carried by a gamete, as each will only receive one of the two homologous chromosomes. (Recall that homologous chromosomes are not identical. They contain different versions of the same genes, and after recombination during crossing over, each gamete will have a unique genetic makeup that has never existed before.)

The randomness in the alignment of recombinant chromosomes at the metaphase plate, coupled with the crossing over events between nonsister chromatids, are responsible for much of the genetic variation in the offspring. To clarify this further, remember that the homologous chromosomes of a sexually reproducing organism are originally inherited as two separate sets, one from each parent. Using humans as an example, one set of 23 chromosomes is present in the egg cell, often called maternal chromosomes because the genetic contributor is often the mother. The other set of 23 chromosomes is contained in the sperm, and the genetic contributor is called a father who provides the paternal chromosomes. Every cell of the multicellular offspring has copies of the original two sets of homologous chromosomes. When the offspring human creates their own gametes through meiosis, the two sets of chromosomes will be rearranged. In prophase I of meiosis, the homologous chromosomes form the tetrads. In metaphase I, these pairs line up at the midway point between the two poles of the cell to form the metaphase plate. Because there is an equal chance that a microtubule fiber will encounter a maternally or paternally inherited chromosome, the arrangement of the tetrads at the metaphase plate is random. Thus, any maternally inherited chromosome may face either pole. Likewise, any paternally inherited chromosome may also face either pole. *The orientation of each tetrad is independent of the orientation of the other 22 tetrads.*

This event— Independent Assortment of homologous chromosomes at the metaphase plate—is the second mechanism that introduces variation into the gametes. In each cell that undergoes meiosis, the arrangement of the tetrads is different. The number of variations is dependent on the number of chromosomes making up a set. There are two possibilities for orientation at the metaphase plate; the possible number of alignments therefore equals 2^n in a diploid cell, where n is the number of chromosomes per haploid set. Humans have 23 chromosome pairs, which results in over eight million (2^{23}) possible genetically-distinct gametes just from the random alignment of chromosomes at the metaphase plate. This number does not include the variability that was previously produced by crossing over between the nonsister chromatids. Given these two mechanisms, it is highly unlikely that any two haploid cells resulting from meiosis will have the same genetic composition (Figure 11.5).

To summarize, meiosis I creates genetically diverse gametes in two ways. First, during prophase I, crossover events between the nonsister chromatids of each homologous pair of chromosomes generate recombinant chromatids with new combinations of maternal and paternal genes. Second, the random assortment of tetrads on the metaphase plate produces unique combinations of maternal and paternal chromosomes that will make their way into the gametes.



This illustration shows that, in a cell with a set of two chromosomes, four possible arrangements of chromosomes can give rise to eight different kinds of gamete. These are the eight possible arrangements of chromosomes that can occur durin...

Figure 11.5 Random, independent assortment during metaphase I is demonstrated by considering a cell with a set of two chromosomes (n = 2). There are two possible homologous chromosome arrangements at the equatorial plane in metaphase I, that are then separated during anaphase I. The total possible number of different gametes is 2^n, where n equals the number of chromosomes in a set. In this example, there are four possible genetic combinations for the gametes. With n = 23 in human cells, there are over eight million possible combinations of paternal and maternal chromosomes. Credit: Rao, A.and Fletcher, S. Department of Biology, Texas A&M University.

Anaphase I

In anaphase I, the microtubules pull the linked chromosomes apart. The sister chromatids remain tightly bound together at the centromere. The proteins holding the synapsed homologous chromosomes are brokendown in anaphase I as the microtubules attached to the fused kinetochores pull the homologous chromosomes apart (Figure 11.6).

Telophase I and Cytokinesis I

In telophase I, the separated chromosomes arrive at opposite poles. The remainder of the typical telophase events occur during this phase. The chromosomes “decondense” and nuclear envelopes form around the separated sets of chromatids produced during telophase I. Then, **cytokinesis**—the physical separation of the cytoplasmic components into two daughter cells—occurs. Cytokinesis separates the cell into two distinct cells via a *cleavage furrow* (constriction of the actin ring that leads to cytoplasmic division).

Two haploid cells are the result of the first meiotic division. The cells are haploid because at each pole, there is just one of each pair of the homologous chromosomes. Therefore, only one full set of the chromosomes is present. This is why the cells are considered haploid—there is only one chromosome set, *even though each chromosome still consists of two sister chromatids.* Recall that sister chromatids are merely duplicates of one of the two homologous chromosomes (except for changes that occurred during crossing over). In meiosis II, these two sister chromatids will separate, creating four haploid daughter cells.

LINK TO LEARNING

Review the process of meiosis, observing how chromosomes align and migrate, at Meiosis: An Interactive Animation .

Meiosis II

There is no Interphase between meiosis I and meiosis II. The two cells produced in meiosis I go directly through the events of meiosis II in synchrony. During meiosis II, the sister chromatids within the two daughter cells separate, forming four new haploid gametes. The mechanics of meiosis II are similar to mitosis, except that each dividing cell has only one set of homologous chromosomes, each with two chromatids.

Prophase II

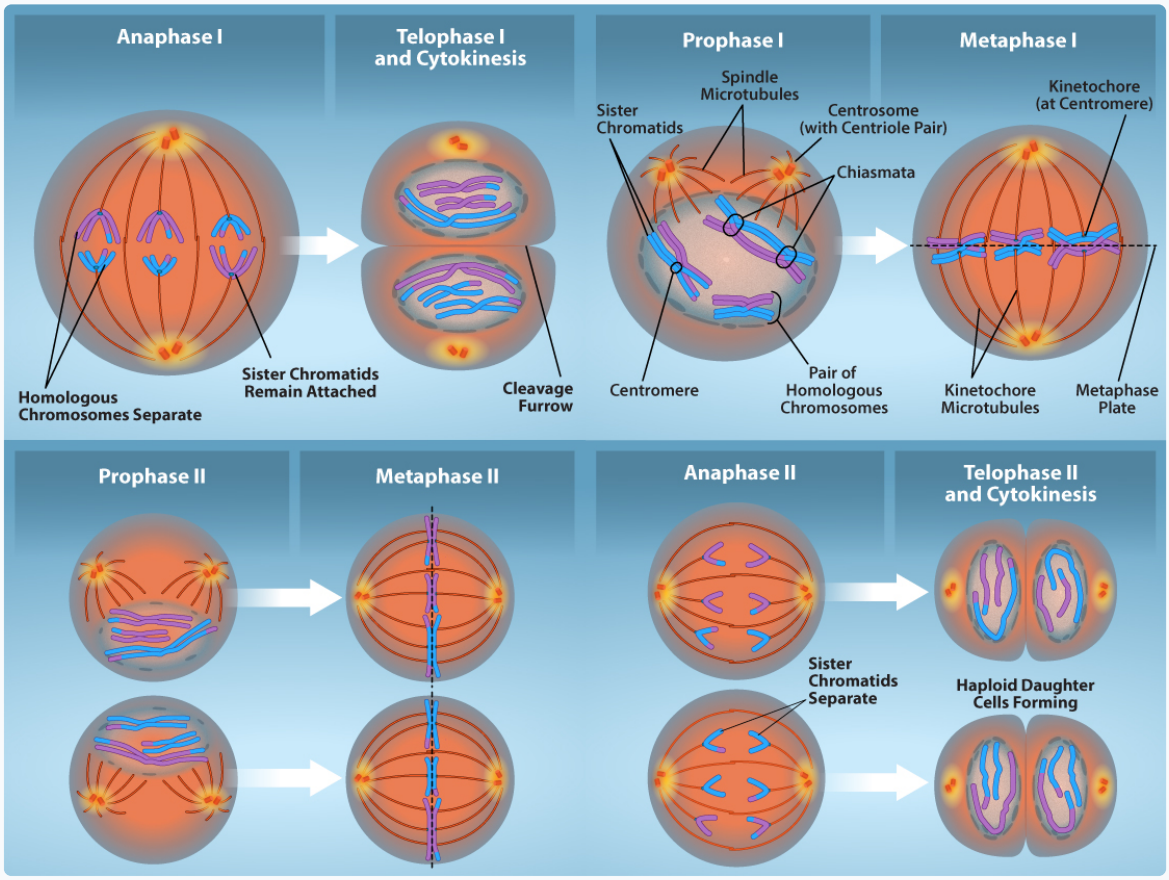
The chromosomes recondense in prophase I, and the nuclear envelope in each cell begins to breakdown until they have completely broken down. Microtubules attach to each sister chromatid at the kinetochore.

Metaphase II

The sister chromatids are maximally condensed and aligned at the equator of the cell such that one sister chromatid of the pair is aligned facing one centrosome and the other sister chromatid of the pair is aligned facing the opposite centrosome.

Anaphase II

The sister chromatids are pulled apart by the kinetochore microtubules and move toward opposite poles.



This illustration compares chromosome alignment in meiosis I and meiosis II. In prophase 1, homologous pairs of chromosomes are held together by chiasmata. In anaphase 1, the homologous pair separates and the connections at the chiasmata ar...

Figure 11.6 The process of chromosome alignment differs between meiosis I and meiosis II. In prophase I, pairs of homologous chromosomes form chiasmata which allow for crossing over events (genetic diversity). These pairs of homologous chromosomes arrange at the metaphase plate in metaphase I. In anaphase I, homologous chromosomes separate. Telophase I and cytokinesis result in haploid cells with 2 sister chromatids of each chromosome. In prophase II, spindle microtubules form and elongate and any nuclear envelope disappears. Sister chromatids arrange at the midpoint of the cells in metaphase II. In anaphase II, the sister chromatids separate. Telophase II and cytokinesis result in haploid cells with a single copy of each chromosome. Credit: Rao, A., Ryan, K., Fletcher, S. and Tag, A. Department of Biology, Texas A&M University.

Telophase II and Cytokinesis II

The chromosomes arrive at opposite poles and begin to decondense. Nuclear envelopes form around the chromosomes. Cytokinesis now separates the two cells into four unique haploid cells. The cells produced are genetically unique because of the random assortment of paternal and maternal homologs and because of the recombination of maternal and paternal segments of chromosomes (with their sets of genes) that occurs during crossover.

Comparing Meiosis and Mitosis

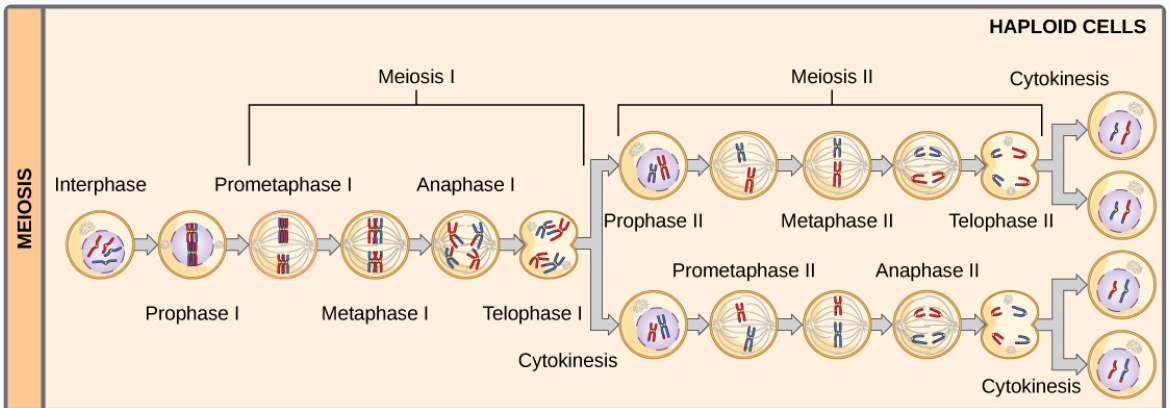
Mitosis and meiosis are both forms of division of the nucleus in eukaryotic cells. They share some similarities, but also exhibit a number of distinct processes that lead to very different outcomes (Figure 11.8). Mitosis is a single nuclear division that results in two nuclei that are usually partitioned into two new cells. The daughter cells produced during mitotic division are genetically identical to the original parent cell. They have the same number

of sets of chromosomes. In contrast, meiosis consists of two nuclear divisions resulting in four daughter cells, called gametes, that are genetically distinct cells. The four gametes produced during meiosis are not genetically identical, and they contain one chromosome set only. This is half the number of chromosome sets of the original cell, which is diploid.

The main differences between mitosis and meiosis occur in meiosis I, which is a very different nuclear division than mitosis. In meiosis I, the homologous chromosome pairs physically meet and synapse. Following this, the chromosomes undergo crossing over between nonsister homologous chromatids. In the end, the homologous chromosomes line up along the metaphase plate as tetrads—with kinetochore fibers from opposite spindle poles attached to each kinetochore of a homolog to form a tetrad. *All of these events occur only in meiosis I.*

Then homologous chromosomes are seprated and move to one pole or the other. After telophase/cytokinesis I the ploidy level has been reduced and both daughter cells are haploid. For this reason, meiosis I is referred to as a reductional division. There is no such reduction in ploidy level during mitosis.

Meiosis II is analogous to a mitotic division. In this case, sister chromatids line up on the metaphase plate during metaphase II. During anaphase II, sister chromatids are separated —they are now daugher chromosomes — one daughter chromosome is pulled toward one pole while the other daughter chromosome is pulled toward the other pole. Meiosis II is not a reduction division because although there are fewer copies of the genome in the resulting cells, there is still one set of chromosomes, as there was at the end of meiosis I.



						OUTCOME
PROCESS	DNA synthesis	Synapsis of homologous chromosomes	Crossover	Homologous chromosomes line up at metaphase plate	Sister chromatids line up at metaphase plate	Number and genetic composition of daughter cells
MEIOSIS	Occurs in S phase of interphase	During prophase I	During prophase I	During metaphase I	During metaphase II	Four haploid cells at the end of meiosis II
MITOSIS	Occurs in S phase of interphase	Does not occur in mitosis	Does not occur in mitosis	Does not occur in mitosis	During metaphase	Two diploid cells at the end of mitosis

This illustration compares meiosis and mitosis. In meiosis, there are two rounds of cell division, whereas there is only one round of cell division in mitosis. In both mitosis and meiosis, DNA synthesis occurs during S phase. Synapsis of ho...

Figure 11.8 Meiosis and mitosis are both preceded by one cycle of DNA replication; however, meiosis includes two nuclear divisions. The four daughter cells resulting from meiosis are haploid and genetically distinct. The daughter cells resulting from mitosis are diploid and identical to the parent cell.

LINK TO LEARNING

Click through the steps of this interactive animation to compare the meiotic process of cell division to that of mitosis at How Cells Divide .

Section 13.4: Instructor Lecture Slide Review (58 Slides Synthesis)

Comprehensive pedagogical synthesis of Dr. Dawson's Chapter 13 lecture presentation (58 slides).

Part A: Modes of Reproduction & Ploidy Foundations (Slides 1–10)

- **Asexual vs. Sexual Reproduction:**
  - **Asexual Reproduction:** A single parent produces genetically identical offspring (clones) via mitosis. Requires less energy and no mate search, but produces zero genetic variation. Mutations are the only source of novelty.
  - **Sexual Reproduction:** Two parents produce specialized haploid gametes (sperm and egg) via meiosis. Fusion during fertilization creates a unique diploid zygote possessing mixed maternal and paternal genomes. Highly favored in changing or unpredictable environments.
- **Ploidy Definitions:**
  - **Diploid (2n):** Cells containing two full sets of chromosomes (one maternal, one paternal). In humans, 2n = 46. Found in somatic body cells.
  - **Haploid (n):** Cells containing a single complete set of chromosomes. In humans, n = 23. Found exclusively in mature gametes.

Part B: The Mechanics of Meiosis I and Meiosis II (Slides 11–44)

- **Overview:** One round of DNA replication in S phase followed by **two consecutive cell divisions** (Meiosis I and Meiosis II), producing **four genetically distinct haploid daughter cells**.
- **Meiosis I (The Reductional Division):** Reduces ploidy from diploid (2n) to haploid (n) by separating homologous chromosomes:
  - **Prophase I:** Chromosomes condense. **Synapsis** occurs: homologous chromosomes pair up precisely gene-for-gene along a protein zipper (the synaptonemal complex) to form **tetrads** (bivalents of 4 chromatids). **Crossing over** occurs: non-sister chromatids exchange reciprocal segments of DNA at physical cross-over points called **chiasmata**, generating **recombinant chromosomes**.
  - **Metaphase I:** Homologous pairs (tetrads) align along the metaphase plate in double-file arrangement. Kinetochores of each homolog are attached to spindle fibers from opposite poles.
  - **Anaphase I:** Homologous chromosomes are pulled apart toward opposite spindle poles by depolymerizing microtubules. *Sister chromatids remain joined at their centromeres!*
  - **Telophase I & Cytokinesis:** Two haploid nuclei form. Each cell now has n chromosomes, but each chromosome still consists of two sister chromatids. No DNA replication occurs between Meiosis I and Meiosis II!
- **Meiosis II (The Equational Division):** Separates sister chromatids, mirroring a mitotic division in haploid cells:
  - **Prophase II:** Spindle apparatus reforms in each of the two daughter cells.
  - **Metaphase II:** Chromosomes align single-file along the metaphase plate. Because of crossing over in Prophase I, the sister chromatids are genetically non-identical.
  - **Anaphase II:** Centromeric cohesins are cleaved; sister chromatids separate into individual chromosomes and move toward opposite poles.
  - **Telophase II & Cytokinesis:** Four distinct nuclei form; cytokinesis divides the cytoplasm, resulting in **four haploid daughter cells**, each with a unique combination of alleles.

Part C: Three Primary Sources of Genetic Variation (Slides 45–50)

Sexual reproduction generates unprecedented genomic diversity through three independent randomizing mechanisms:

1. **Crossing Over (Prophase I):** Creates novel allele combinations by physically splicing maternal and paternal chromatid fragments together, generating recombinant chromosomes. Typically 1 to 3 crossover events occur per chromosome pair in humans.
2. **Independent Assortment of Chromosomes (Metaphase I):** Each homologous pair aligns at the metaphase plate with random maternal/paternal orientation, completely independent of all other pairs. The number of possible daughter cell combinations resulting from independent assortment is $\mathbf{2^n}$, where n is the haploid number. In humans (n = 23), $2^{23} \approx$ **8.4 million distinct gametic combinations** per individual!
3. **Random Fertilization:** Any one of an individual's 8.4 million possible sperm can fertilize any one of an individual's 8.4 million possible eggs:

   $$\mathbf{2^{23} \times 2^{23} \approx 70 \text{ Trillion Unique Zygotic Combinations!}}$$

   (Excluding crossing over, which increases diversity exponentially toward infinity).

Part D: Meiotic Errors, Nondisjunction & Chromosomal Aberrations (Slides 51–58)

- **Nondisjunction:** The failure of homologous chromosomes to separate normally during Meiosis I, or the failure of sister chromatids to separate during Meiosis II:
  - **Meiosis I Nondisjunction:** All four resulting gametes are abnormal (two gametes with n + 1; two gametes with n - 1; 100% defective).
  - **Meiosis II Nondisjunction:** Sister chromatids fail to separate in one cell; results in two normal gametes (n, n), one with n + 1, and one with n - 1 (50% defective).
- **Aneuploidy:** An abnormal number of chromosomes in a zygote following fertilization by a nondisjunction gamete:
  - **Monosomy (2n - 1):** Missing one chromosome. Example: **Turner Syndrome (45, X0)** — the only viable human monosomy. Phenotypically female, sterile, short stature.
  - **Trisomy (2n + 1):** Having three copies of a chromosome.
    - **Trisomy 21 (Down Syndrome):** 47 chromosomes (+21). Most survivable autosomal trisomy; developmental delay, heart defects, life expectancy ~60 years.
    - **Trisomy 18 (Edwards Syndrome):** 47 chromosomes (+18). Severe organ and craniofacial malformations; rarely survive beyond infancy.
    - **Trisomy 13 (Patau Syndrome):** 47 chromosomes (+13). Severe brain, ocular, and cardiac defects; typically fatal within days/weeks.
  - **Sex Chromosome Trisomies:**
    - **Klinefelter Syndrome (47, XXY):** Phenotypically male; reduced testosterone, partial breast enlargement, sterile.
    - **Triple X Syndrome (47, XXX):** Phenotypically female; often asymptomatic due to Barr body X-inactivation.
    - **Jacobs Syndrome (47, XYY):** Phenotypically male; taller than average stature, fertile.
- **Structural Alterations in Chromosomes:**
  - **Deletion:** Loss of a chromosomal segment.
  - **Duplication:** Repetition of a chromosomal segment from unequal crossing over.
  - **Inversion:** Reversal of a segment within the chromosome.
  - **Translocation:** Movement of a chromosomal segment to a non-homologous chromosome.

Section 13.5: Chapter 13 Official Vocabulary & Conceptual Foundations

Key terms transcribed from the official course glossary with precise biological definitions.

Synapsis
The physical pairing and lateral union of homologous chromosomes along a proteinaceous synaptonemal complex during Prophase I of Meiosis, forming tetrads.

Crossing Over
The reciprocal exchange of genetic material between non-sister chromatids of homologous chromosomes during Prophase I, generating recombinant chromosomes.

Chiasma (plural: Chiasmata)
The X-shaped microscopic physical point of attachment where crossing over and genetic exchange have occurred between homologous chromatids.

Recombinant Chromosome
A chromosome that has undergone meiotic crossing over, carrying a novel combination of maternal and paternal DNA alleles.

Law of Independent Assortment
Mendel's second law; states that each pair of homologous chromosomes aligns and segregates independently of all other pairs during Metaphase I / Anaphase I.

Nondisjunction
An error in meiosis (or mitosis) in which homologous chromosomes fail to separate in Meiosis I, or sister chromatids fail to separate in Meiosis II.

Aneuploidy
A chromosomal aberration characterized by an abnormal number of chromosomes (having extra or missing chromosomes relative to the wild-type 2n).

Monoploidy / Monosomy
A condition where a diploid organism is missing one chromosome of a homologous pair (2n - 1), such as Turner Syndrome (X0).

Polyploidy / Trisomy
Trisomy refers to having three copies of a specific chromosome (2n + 1). Polyploidy refers to possessing three or more entire complete chromosome sets (3n, 4n).

Tetrad (Bivalent)
A four-part structure consisting of two paired homologous chromosomes (each with 2 sister chromatids) formed during Prophase I synapsis.

Section 13.6: Official Course Practice Problems & Verified Solutions

Official examination practice problems from Chapters 12 and 13 Practice Questions.docx with complete answers and explanations.

1. Which of the following occurs in meiosis but NOT mitosis?

A) Sister chromatids separate • **B) Homologous chromosomes synapse (Correct)** • C) Chromosomes are replicated • D) Daughter cells are produced

Rationale: Synapsis of homologous chromosomes and crossing over occur exclusively during Prophase I of meiosis. Mitosis involves no pairing of homologs.

2. How is metaphase of Meiosis I different from metaphase of Meiosis II?

Answer: In **Metaphase I**, homologous chromosome pairs (tetrads) align double-file along the metaphase plate. In **Metaphase II**, individual chromosomes align single-file along the metaphase plate (with sister chromatids facing opposite poles, exactly like mitosis).

3. In a dog karyotype, chromosomes are arranged in 39 pairs (2n = 78). If we choose one homologous pair, such as pair 14, which of the following do the two chromosomes of the pair have in common?

A) Length and position of the centromere only • B) Length, centromere position, and staining pattern only • **C) Length, centromere position, staining pattern, and types of genes (Correct)** • D) They have nothing in common

Rationale: Homologous chromosomes share morphology, centromere location, banding pattern, and identical gene loci (though they may carry different allele variants).

4. Which of the following is TRUE at the end of Meiosis I?

A) The chromosomes are no longer comprised of sister chromatids • B) The chromosome number per cell remains the same as parent cell • **C) The cells are haploid (Correct)** • D) Four daughter cells have been formed

Rationale: Meiosis I separates homologous pairs, reducing the ploidy from diploid (2n) to haploid (n). Each daughter cell contains n chromosomes, each still consisting of two sister chromatids.

5. At what phase of meiosis does crossing over occur?

A) Meiosis II • B) Anaphase II • **C) Prophase I (Correct)** • D) Prophase II

Rationale: Synapsis and chiasma formation occur strictly during Prophase I between non-sister chromatids of homologous pairs.

6. Sister chromatids separate at which phase of Meiosis?

A) Anaphase I (III) • B) Telophase I (IV) • C) Prophase II (V) • **D) Anaphase II (VII) (Correct)**

Rationale: Homologous chromosomes separate in Anaphase I; sister chromatids separate in Anaphase II.

7. The cells are considered haploid at the end of Meiosis I and Meiosis II. True/False:

A) True (Correct) • B) False

Rationale: Cells become haploid (n) at the conclusion of Meiosis I when homologs are segregated, and remain haploid (n) throughout Meiosis II as sister chromatids separate.

8. During which of the following processes do sister chromatids align along the metaphase plate of a cell? (Choose all that apply)

A) Metaphase of mitosis & C) Metaphase II of meiosis (Correct) • B) Metaphase I of meiosis • D) None of the above

Rationale: In mitotic metaphase and meiotic metaphase II, individual chromosomes align single-file with sister chromatids facing opposite poles. In metaphase I, tetrads align double-file.

9. After a sperm fertilizes an egg, what must the newly formed diploid cell do?

A) Keep chromosomes separate • **B) Begin to divide via mitosis (Correct)** • C) Join homologous chromosomes • D) Join with another newly fertilized cell

Rationale: The unicellular zygote undergoes repeated rounds of mitotic division and differentiation to develop into a multicellular embryo.

10. Explain why ovary and testes cells must undergo meiosis rather than mitosis:

Answer: Gametes must be haploid (n) so that upon fertilization (sperm + egg), the resulting single-celled zygote restores the diploid chromosome number (2n) characteristic of the species. If gametes were produced by mitosis (2n), chromosome numbers would double exponentially every generation (4n, 8n, 16n), causing fatal developmental breakdown.

11. A cell is 2n = 28. How many sister chromatids are present in EACH cell in Metaphase II of Meiosis II?

A) 0 • B) 14 • **C) 28 (Correct)** • D) 56

Rationale: At Meiosis I, the 28 duplicated chromosomes are divided into two cells, each containing n = 14 duplicated chromosomes. Because each chromosome consists of 2 sister chromatids: 14 × 2 = 28 sister chromatids per cell.

12. A cell is 2n = 30. How many tetrads are present during Prophase I of Meiosis I?

A) 15 (Correct) • B) 30 • C) 45 • D) 60

Rationale: A tetrad consists of one homologous pair (2 chromosomes, 4 chromatids). With 2n = 30 chromosomes, there are 30 / 2 = 15 homologous pairs, and therefore exactly 15 tetrads.

Appendix: Exam 4 Focus Specifications & Meiosis Ploidy Matrix

Official curriculum requirements from Exam 4 Study Guide 2026.docx covering meiotic chromosome segregation and nondisjunction mechanisms.

Summary Matrix: Chromosome & Chromatid Accounting Through Meiosis (Example: 2n = 46)

Stage	Ploidy Level	Chromosomes	Sister Chromatids	Arrangement & Event
G1 Phase (Pre-Meiotic)	Diploid (2n)	46	0	Unreplicated germ cell chromatin
Prophase I / Metaphase I	Diploid (2n)	46	92	23 tetrads align double-file; crossing over
Anaphase I	Diploid (2n)	46	92	Homologs separate; sister chromatids stay joined
Telophase I (Per Cell)	**Haploid (n)**	**23**	**46**	Two haploid cells; chromosomes still duplicated
Prophase II / Metaphase II	Haploid (n)	23 per cell	46 per cell	Chromosomes align single-file at equator
Anaphase II (Per Cell)	Diploid transiently (2n)	46 per cell	0	Centromeres cleave; chromatids separate
Telophase II (Per Gamete)	**Haploid (n)**	**23**	**0**	4 unique haploid gametes produced!

Direct Comparison: Mitosis vs. Meiosis Summary

Property	Mitosis	Meiosis
Site of Occurrence	Somatic body cells	Germline cells (ovaries / testes)
Number of Divisions	One division (M phase)	Two divisions (Meiosis I & II)
Synapsis & Crossing Over	Does not occur	Occurs in Prophase I (chiasmata)
Daughter Cell Number	Two daughter cells	Four daughter cells
Daughter Cell Ploidy	Diploid (2n) — identical to parent	Haploid (n) — half of parent
Genetic Identity	Genetically identical clones	Genetically non-identical (diverse)
Biological Function	Growth, tissue repair, asexual reproduction	Production of gametes for sexual reproduction